

SANTA FE, Argentina

WMO: 873710

Lat: 31.7078S

Lon: 60.8053W

Elev: 18

StdP: 101.11

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.9	2.9	-3.1	2.9	6.5	-1.4	3.4	6.7	16.2	12.6	14.5	14.8	1.0	230	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	35.1	24.2	33.6	23.5	32.2	23.0	26.4	32.0	25.5	30.9	24.7	29.8	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.0	29.8	24.0	18.9	28.8	23.1	17.9	27.9	82.6	32.0	78.7	31.0	75.1	30.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.0	10.3	-2.3	38.6	1.7	1.4	-3.5	39.6	-4.5	40.5	-5.5	41.3	-6.8	42.3		
(5)			-2.9	28.0	1.6	0.9	-4.1	28.7	-5.0	29.2	-5.9	29.8	-7.1	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	25.8	24.6	22.6	19.2	15.9	12.9	12.2	14.2	16.5	19.4	22.3	24.5	(6)
(7)		DBStd	5.89	2.72	2.96	3.12	3.68	3.96	4.00	4.47	4.73	4.16	3.54	3.13	2.96	(7)
(8)		HDD10.0	52	0	0	0	0	2	13	24	10	2	0	0	0	(8)
(9)		HDD18.3	769	0	1	4	32	97	171	199	149	83	28	5	1	(9)
(10)		CDD10.0	3389	491	408	390	276	187	100	93	139	196	293	368	449	(10)
(11)		CDD18.3	1066	233	176	136	58	23	8	9	20	27	62	123	191	(11)
(12)		CDH23.3	9404	2433	1586	1062	362	104	24	44	159	231	476	1062	1861	(12)
(13)		CDH26.7	3553	1072	612	348	89	19	3	9	50	71	142	369	768	(13)
(14)	Wind	WSAvg	3.4	2.9	2.9	2.8	2.9	2.9	3.4	3.6	4.1	4.1	4.1	4.0	3.5	(14)
(15)	Precipitation	PrecAvg	991	110	106	143	102	48	31	27	34	53	100	118	107	(15)
(16)		PrecMax	1654	276	415	408	392	154	184	81	196	218	313	290	331	(16)
(17)		PrecMin	496	4	3	20	6	1	0	0	0	4	28	14	11	(17)
(18)		PrecStd	229	78	79	82	79	42	37	21	39	48	60	65	73	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	35.7	34.1	31.8	29.1	26.3	27.8	31.6	33.0	33.6	34.7	36.5	(19)
(20)			MCWB	24.9	25.0	24.8	23.6	22.0	21.0	19.8	21.7	22.3	22.7	23.2	23.6	(20)
(21)		2%	DB	35.1	33.7	31.9	29.1	26.0	23.0	23.8	27.4	28.4	30.1	32.2	34.3	(21)
(22)			MCWB	24.4	24.4	23.4	21.8	20.1	19.1	18.6	19.6	19.9	21.4	21.7	23.5	(22)
(23)		5%	DB	33.3	32.0	30.1	27.1	23.9	20.9	21.0	24.4	25.9	27.9	30.5	32.6	(23)
(24)			MCWB	23.6	24.0	22.7	21.0	19.4	17.5	16.6	17.9	18.7	20.3	21.0	22.7	(24)
(25)		10%	DB	31.9	30.5	28.5	25.2	21.9	18.9	19.0	21.8	23.6	25.9	28.7	30.8	(25)
(26)			MCWB	23.2	23.3	21.9	20.0	18.0	16.0	15.4	16.4	17.1	19.1	20.5	22.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.4	26.8	25.0	22.8	21.7	21.0	22.3	23.3	25.1	25.2	26.8	(27)
(28)			MCDWB	33.8	32.6	31.4	29.8	27.2	25.0	25.9	29.0	30.0	31.5	31.6	33.5	(28)
(29)		2%	WB	26.2	26.2	25.1	23.6	21.3	19.9	19.1	20.4	21.1	22.9	23.5	25.5	(29)
(30)			MCDWB	32.2	31.1	29.3	27.3	24.6	22.2	22.8	26.3	27.0	27.6	29.2	31.2	(30)
(31)		5%	WB	25.3	25.2	23.8	22.2	20.1	18.3	17.6	19.0	19.5	21.5	22.5	24.4	(31)
(32)			MCDWB	31.1	29.8	28.0	25.2	23.0	20.4	20.5	23.6	24.2	26.0	27.9	29.7	(32)
(33)		10%	WB	24.4	24.2	22.8	21.1	18.8	16.7	16.0	17.1	18.0	20.2	21.5	23.3	(33)
(34)			MCDWB	29.8	28.9	27.1	24.1	21.3	18.4	18.3	20.8	22.3	24.6	26.7	28.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	9.6	9.9	9.6	9.0	9.3	10.0	10.8	10.8	10.1	10.7	10.5	(35)
(36)			MCDBR	12.6	11.8	11.6	12.0	10.7	10.4	11.6	13.4	13.9	12.9	13.3	13.2	(36)
(37)			MCWBR	4.9	4.9	5.0	5.9	5.6	6.2	6.5	6.9	6.9	6.1	5.5	5.4	(37)
(38)		5% WB	MCDWB	10.4	9.7	9.9	9.3	8.8	8.1	9.8	11.9	11.4	10.6	10.7	10.5	(38)
(39)			MCWBR	5.1	5.1	5.2	5.6	5.1	5.4	6.0	6.5	6.5	6.1	5.5	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.403	0.393	0.377	0.385	0.364	0.355	0.359	0.427	0.490	0.433	0.397	0.400	(40)	
(41)		taud	2.374	2.418	2.455	2.399	2.414	2.430	2.392	2.151	1.982	2.221	2.347	2.366	(41)	
(42)		Ebn at Noon	937	927	907	841	806	788	802	780	782	879	936	943	(42)	
(43)		Edh at Noon	130	120	109	104	91	85	92	130	172	145	133	132	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.15	6.15	5.27	3.88	2.89	2.50	2.79	3.71	4.80	5.69	6.91	7.24	(44)	
(45)		RadStd	0.46	0.46	0.42	0.49	0.25	0.28	0.29	0.33	0.42	0.46	0.39	0.42	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page